Sri Lanka Institute of Information Technology

Faculty of Computing

IT2011 - Artificial Intelligence and Machine Learning

Lab 05 - Introduction to Generative AI

Submission: Exercises 2 to 6

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Exercise 4: OpenAI API in Python

Baseline test (physics assistant)

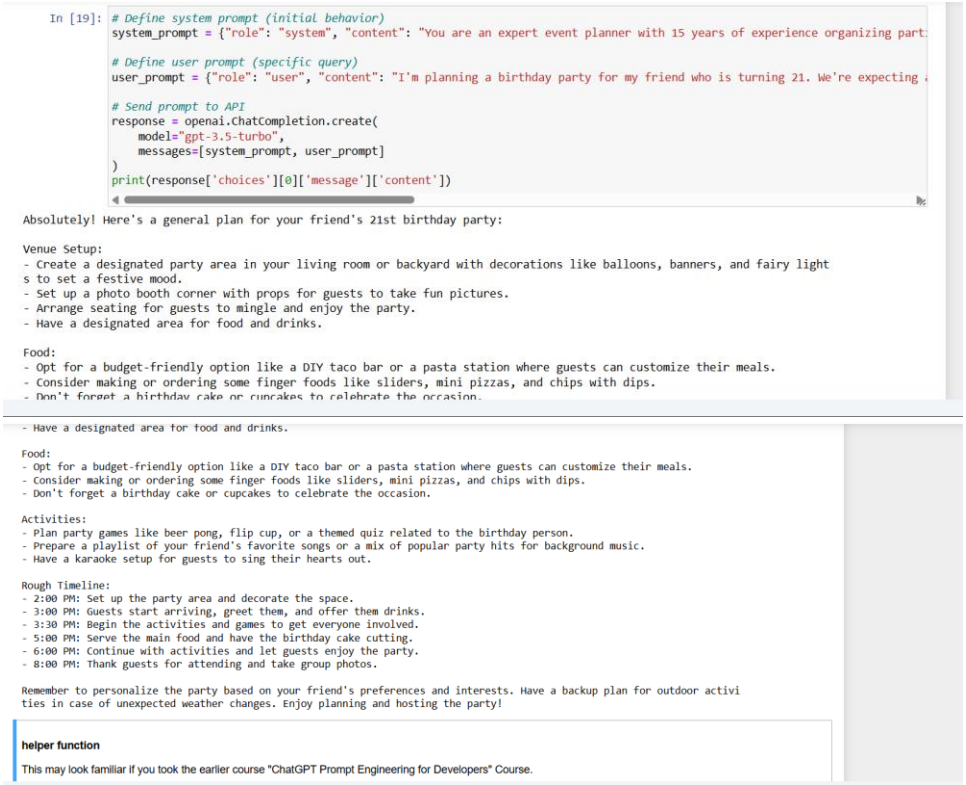
```
openai.api_key  = os.environ['OPENAI_API_KEY']

In [17]: # Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You are a helpful assistant with deep knowledge in physics."}

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "Explain how black holes are formed."}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

```
Black holes are formed when a massive star runs out of fuel and collapses in on itself during a supernova explosion. As t
he star collapses, the outer layers are blasted into space while the core collapses under its own gravity. If the core ha
s a mass greater than about three times that of the Sun, it will collapse into a black hole.

The core of the star becomes so dense that its gravity becomes incredibly strong, causing it to collapse to a point of in
finite density called a singularity. This singularity is surrounded by an event horizon, which is the point of no return
for anything that crosses it. Once an object or light crosses the event horizon, it can never escape the gravitational pu
ll of the black hole.

Black holes come in different sizes, with stellar-mass black holes being formed from the collapse of massive stars, and s
upermassive black holes found at the centers of galaxies, with masses millions or even billions of times that of the Sun.
```

Event Planner Chatbot

System: "You are an expert event planner..."

User: "I'm planning a birthday party for my friend turning 21, 20 guests, $200 budget, at home. Give a full plan."



```
In [19]: # Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You are an expert event planner with 15 years of experience organizing part

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "I'm planning a birthday party for my friend who is turning 21. We're expecting

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

```
Absolutely! Here's a general plan for your friend's 21st birthday party:

Venue Setup:
- Create a designated party area in your living room or backyard with decorations like balloons, banners, and fairy light
s to set a festive mood.
- Set up a photo booth corner with props for guests to take fun pictures.
- Arrange seating for guests to mingle and enjoy the party.
- Have a designated area for food and drinks.

Food:
- Opt for a budget-friendly option like a DIY taco bar or a pasta station where guests can customize their meals.
- Consider making or ordering some finger foods like sliders, mini pizzas, and chips with dips.
- Don't forget a birthday cake or cupcakes to celebrate the occasion.

Activities:
- Plan party games like beer pong, flip cup, or a themed quiz related to the birthday person.
- Prepare a playlist of your friend's favorite songs or a mix of popular party hits for background music.
- Have a karaoke setup for guests to sing their hearts out.

Rough Timeline:
- 2:00 PM: Set up the party area and decorate the space.
- 3:00 PM: Guests start arriving, greet them, and offer them drinks.
- 3:30 PM: Begin the activities and games to get everyone involved.
- 5:00 PM: Serve the main food and have the birthday cake cutting.
- 6:00 PM: Continue with activities and let guests enjoy the party.
- 8:00 PM: Thank guests for attending and take group photos.

Remember to personalize the party based on your friend's preferences and interests. Have a backup plan for outdoor activi
ties in case of unexpected weather changes. Enjoy planning and hosting the party!
```

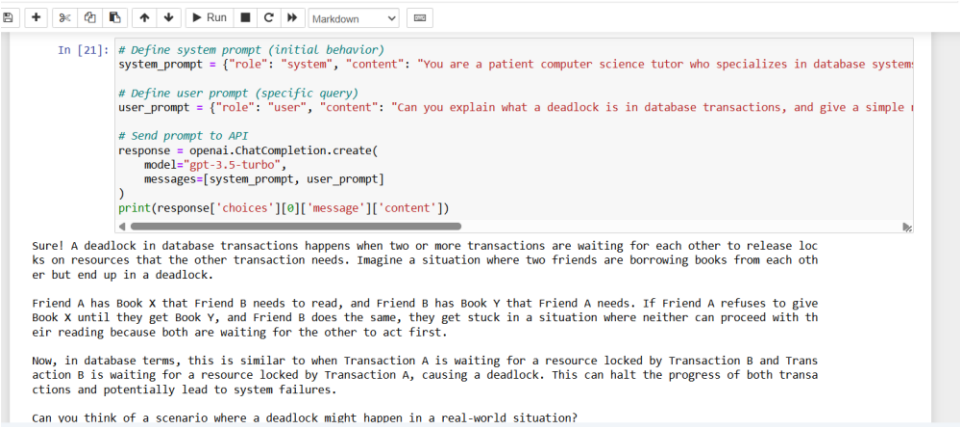
helper function

This may look familiar if you took the earlier course "ChatGPT Prompt Engineering for Developers" Course.

Unique Prompt: Database Systems Tutor

System: "You are a patient CS tutor specializing in database systems..."

User: "Explain what a deadlock is in database transactions, with a real-world analogy."



Changing the system prompt clearly changed the model's persona and output style across all three runs, while the underlying API call stayed identical.